DATA SHEET

RTARIUM

A three-dimensional basalt mesh for advanced concrete applications: two basalt mesh faces, connected by vertical fibres to form a 3D structure.

Non-corrosive · **Alkali-resistant** · **Three-dimensional**

Mesh · Geometry

Structure	**3D woven spacer** Two faces connected by vertical fibres
Thickness options	**3 / 6 / 10 / 15 mm**
Sheet length	**2.4 / 3.4 / 5.8 m**
Sheet width	**1,220 mm**
Surface density	**220 – 320 g/m²** ±10 g/m²

Mesh · Performance

Tensile strength	**> 50 kN/m** Per metre width of mesh
Elastic modulus	**79,600 MPa**
Elongation at break	**3.4 %**

Mesh · Resistance

Sustained temperature	**820 °C**
Melting point	**1,460 – 1,500 °C**
Corrosion	**Non-corrosive**
Alkali	**High resistance**

Impregnated basalt yarn

Tensile strength	**2.62 GPa** Above steel, aircraft aluminium and titanium alloy; below carbon fibre.
Elastic modulus	**79.6 GPa** Stiffer than E-glass and AR-glass fibre.
Elongation at break	**3.40 %** More stretch before it breaks than glass or steel fibre, which helps against cracking and shrinkage.

Tensile strength, MPa

Structural steel	**550**
High-strength aircraft aluminium	**570**
Titanium alloy	**1,200**
RTARIUM yarn	**2,620**
Carbon fibre	**3,500**

Concrete reinforced with RTARIUM

Axial tensile strength	**19.7 MPa** Normal concrete reaches 2 to 5 MPa.
Elastic-limit tensile strength	**18.3 MPa** Up to this stress the matrix behaves elastically; above it, the first internal micro-cracks form.
Tensile elastic modulus	**64 GPa** A 20 mm GFRC is around 25 GPa; UHPC around 50 GPa.

Steel fibres swapped for RTARIUM in UHPC

Tensile strength	**16.2 → 20.9 MPa** +29 %
Flexural strength	**28.6 → 50.6 MPa** +77 %

Figures describe the tested materials and configuration. Performance in a particular element depends on its mix, thickness and detailing. Applications are subject to project-specific engineering and validation.